

Suhail Siddiqui

Python Backend Developer | Data Engineer | AI Integration Engineer

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Professional Summary

Python Backend Developer and Data Engineer with 2+ years of production experience building scalable REST APIs, real-time AI systems, cloud data pipelines, and IoT-integrated backend services. Hands-on experience with AWS, GCP, Azure, Snowflake, IICS, and SSMS across enterprise and SaaS projects. Published researcher in IoT and AI domains through Springer and IJACEN (2025). Currently expanding expertise in LangChain, RAG pipelines, LLM integration, and Agentic AI.

Professional Experience

Full Stack Python Developer / Data Engineer | Natlov Technologies Pvt. Ltd. – Remote Jul 2025 – Present

- Managed and enhanced Columbia University's web platform, resolving GitLab CI/CD tickets and shipping feature improvements using Angular and Flask.
- Developed scalable web applications with Angular, Flask, REST API integration, dynamic tables, search, pagination, and form validation.
- Supported the Ascension project using GCP, Google Cloud Console, Looker dashboards, and Azure data pipelines for cloud-based reporting infrastructure.
- Built and maintained Looker dashboards for business intelligence and operational reporting on GCP.
- Worked on the Toro project using Snowflake and DBT for data transformation and pipeline management.
- Serve as the main developer on the CareFirst project, using SSMS, IICS (Informatica), Snowflake, and MDM for enterprise healthcare data management.

Backend / Integration Engineer (Freelance) | TheTechRavens – Remote Jul 2024 – Jun 2025

- Engineered production Django REST APIs for the JiveBird digital gifting SaaS platform, integrating Stripe, PayPal, Vonage SMS, HubSpot CRM, and AWS services including S3, Lambda, and RDS.
- Built Python/Zeep middleware connecting a React frontend to legacy SOAP webservice for Capital Car Rental, with PostgreSQL caching to eliminate query bottlenecks.
- Maintained and delivered responsive UI enhancements across 9+ Private Jet Nation aviation portals.
- Designed multi-tenant backend systems with JWT authentication, RBAC, and microservices-style API architecture.

Python Django Developer | Skyview Smart Solutions – Lucknow, India Dec 2024 – May 2025

- Built a production-grade real-time AI Virtual Try-On system integrating AI camera capture, computer vision overlay, and WebSocket (Socket.IO) QR-based device pairing; deployed on IIS using ASGI/WSGI.
- Developed a secure Django/DRF backend with JWT authentication, role-based access control, validation, and structured error handling.
- Integrated AI camera capture and overlay features into a live production system serving real users.

Python Django Intern | Skyview Smart Solutions – Lucknow, India Jan 2024 – Jun 2024

- Built an IoT-based smart farming backend with a real-time Arduino/ESP8266 sensor ingestion pipeline for automated agricultural decision-making.
- Designed automation logic, REST APIs, role-based dashboards, and real-time Chart.js visualizations.
- Improved PostgreSQL performance through query tuning, indexing, and caching strategies.

Technical Skills

- **Languages:** Python, Java (Basic), C (Basic), SQL, JavaScript
- **Backend:** Django, Flask, Django REST Framework, FastAPI (Learning), Socket.IO, REST APIs, JWT, RBAC
- **Data Engineering:** Snowflake, DBT, IICS (Informatica), SSMS, MDM, SQL, ETL Pipelines, Looker
- **Cloud & DevOps:** AWS (S3, Lambda, RDS, EC2), GCP (Google Cloud Console, Looker), Azure Pipelines, Docker (Familiar), GitLab CI/CD, GitHub Actions
- **Databases:** PostgreSQL, MongoDB, MySQL, SQLite, Firebase Realtime Database, Snowflake
- **AI / GenAI:** OpenAI GPT APIs, LLM Integration, Prompt Engineering, RAG, FAISS, ChromaDB, LangChain (Basics), Agentic AI (Learning)
- **Frontend:** HTML, CSS, Angular, Tailwind CSS, Bootstrap, jQuery, Chart.js, JavaScript
- **Tools & Deployment:** Git, GitHub, GitLab, IIS, ASGI, WSGI, Postman, VS Code

Key Projects

Retail Shop POS & Inventory Management System

- Built an end-to-end checkout workflow using transaction.atomic, creating SoldCustomer/SoldItem records and marking inventory as sold with transactional safety.
- Implemented resilient inventory lookup using inventory ID, IMEI1, IMEI2, and barcode, including fixes for filtering edge cases.
- Integrated the OpenAI API for voice-to-form extraction, converting unstructured speech into structured product fields.
- Added invoice retrieval, sold-item history, customer purchase history, and administrative password reset endpoints.

Tech: Python, Django 5.x, DRF, PostgreSQL, JWT, OpenAI API, Render

JiveBird – Digital Gifting SaaS Platform

Live Site

- Engineered core Django REST APIs for live payments, communications, and CRM integrations using Stripe, PayPal, Vonage, and HubSpot.
- Supported AWS serverless infrastructure using S3, Lambda, RDS, and PostgreSQL.

Tech: Python, Django REST, PostgreSQL, AWS, Third-party APIs

Smart Meter – IoT Real-Time Energy Monitoring System

GitHub

- Integrated Arduino/ESP8266 sensors (ZMCT101C and ZMPT103B) for live energy ingestion with synchronized voltage-current power calculations.
- Built a Firebase and MongoDB data pipeline and an RBAC dashboard using Tailwind CSS and JavaScript visualizations.
- Published the work through Springer and IJACEN in 2025.

Tech: Python, Django, Firebase, MongoDB, ESP8266, Arduino, Tailwind CSS

Capital Car Rental – SOAP/REST Middleware

- Engineered middleware connecting a React frontend with legacy Rentway SOAP webservice using Python Zeep.
- Added a local PostgreSQL caching layer to optimize API responses and prevent repeated query bottlenecks.

Tech: Python, Django REST, SOAP/Zeep, PostgreSQL, AWS

Research Publications

Real-Time Monitoring and Data Management for Smart Meter using IoT and Cloud Technologies

2025

Suhail Siddiqui, Mohd Faraz Kalim, Saleha Mariyam, Kavita Agarwal. Proceedings of the International Conference on Data Science and Artificial Intelligence (DSAI 2025), Springer. **Published**

Behavioral Nudges and Predictive Analytics: Enhancing Energy Conservation Through IoT-Enabled Smart Metering

2025

Suhail Siddiqui, Mohd Faraz Kalim, Saleha Mariyam, Kavita Agarwal. International Journal of Advance Computational Engineering and Networking (IJACEN). **Published**

Education

B.Tech – Computer Science Engineering (Data Science & AI, IBM Specialization)

2021 – 2025

Integral University, Lucknow | CGPA: 7.2

Certifications

- Cloud Computing Fundamentals – IBM SkillsBuild
- Rapid Development for AI – IBM SkillsBuild
- Cybersecurity Fundamentals – IBM SkillsBuild
- IBM Cognos Analytics Reporting Essentials – IBM SkillsBuild
- Industry 4.0 & Industrial IoT – NPTEL (SWAYAM)
- Introduction to Internet of Things (IoT) – NPTEL (SWAYAM)
- Postman API Fundamentals – Student Expert – Postman
- Web Development with Python – Softlew Technologies

Languages

English: Intermediate | Hindi: Native | Urdu: Fluent